MIL-F-28675A
16 November 1979
SUPERSEDING
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10 December 1974

MILITARY SPECIFICATION

FITTINGS, CONDUIT, EXPLOSION PROOF (COUPLINGS, SEALS, UNIONS, ELBOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers explosion-proof electrical conduit fittings including conduit seals, flexible couplings, reducers, unions, drain and breather valves, inspection seals, drain seals, and elbows.

1.2 Classification.

1.2.1 Types, classes, and styles. The fittings shall be one of the following types, classes, and styles as specified (see 6.2):

Type I - Conduit seals.

Class 1 - Vertical.
Class 2 - Vertical/horizontal.
Class 3 - Multi-angular.

Style A - Female hub, top and bottom.
Style B - Male hub, top and female, bottom.
Style C - Female hub, top and male, bottom.
Style D - Elbow, female hub, top and bottom.

Type II - Flexible couplings.

Class 1 - Female with removable male nipple and female union.
Class 2 - Female with removable male nipples, both ends.
Class 3 - Male threaded end fittings.
Class 4 - Male threaded end fitting on one end. Female union on the other.

Type III - Reducer.
Type IV - Unions.

Class 5 - Male.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self addressed Standardization Document Improvment Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5975

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Class 6 - Female.
Class 7 - Angle, 90° elbow.
Class 8 - Universal, male.
Class 9 - Universal, female.

Type V - Drain and breather valve.
Type VI - Inspection seal.
Type VII - Drain seal.
Type VIII - Elbows.

Class 10 - 90° male.
Class 11 - 90° female.
Class 12 - 45° female.
Class 13 - 90° male and female.
Class 14 - Flexible.

1.2.2 Size. Fitting sizes shall be 1/8, 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, or 6 inch sizes as specified (see 6.2). Fitting sizes as related to type, class, and style shall be as shown in Tables I through VII.

TABLE I. Type I conduit seals.

Style (inches)	Class 1 sizes (inches)	Class 2 sizes (inches)	Class 3 sizes (inches)
A	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6	
B	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6	
D		3/4	
E			1/2, 3/4, 1, 1-1/4, 1-1/2, 2/ 2-1/2, 3
F			2, 2-1/2,3

TABLE II. Type II flexible couplings.

Flexible length (inches)	All classes Size (inches)
4	1/2, 3/4
6	1/2, 3/4, 1
8	1/2, 3/4, 1
10	1/2, 3/4, 1
12	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
15	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
18	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
21	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
24	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
27	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
30	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
33	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
36	1/2, 3/4, 1, 1-1/4, 1-1/2, 2

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TABLE III. Type III reducers.

Female thread Size (inches)	Male thread Size (inches)	Female thread Size (inches)	Male thread Size (inches)
1/2	1/8	2-1/2	1
1/2	1/4	2-1/2	1-1/4
1/2	3/8	2-1/2	1-1/2
3/4	1/2	2-1/2	2
1	1/2	3	1
1	3/4	3	1-1/4
1-1/4	1/2	3	1-1/2
1-1/4	3/4	3	2
1-1/4	1	3	2-1/2
1-1/2	1/2	3-1/2	2
1-1/2	3/4	3-1/2	2-1/2
1-1/2	1	3-1/2	3
1-1/2	1-1/4	4	2
2	1/2	4	2-1/2
2	3/4	4	3
2	1	4	3-1/2
2	1-1/4	5	4
2	1-1/2	6	5

TABLE IV. Type IV unions.

Class	Size (inches)
5	1/2, 3/4 to 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6
6	1/2, 3/4 to 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6
7	1/2 to 1/2, 1/2 to 3/4, 3/4 to 1/2, 3/4 to 3/4
8	1/2, 3/4, 1
9	1/2, 3/4, 1

TABLE V. Type V drain and breather valves.

Size (inches)
1/4
3/8
1/2 (straight)
1/2 (90 degree elbow)

TABLE VI. Type VI inspection seals and type VII drain seals.

Type	Hub size (inches)
VI	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
VII	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4

TABLE VII. Type VIII elbows.

Class	Size (inches)
10	1/2, 3/4, 1, 1-1/4
11	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2
12	1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4
13	1/2, 3/4, 1, 1-1/4, 1-1/2, 2
14	1/2, 3/4, 1

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* 1.2.3 Design installation use. When required for a specific application in an existing or planned installation, fittings shall be designated as to use as specified (see 6.2).

Use (a) - Vertical only.
Use (b) - Horizontal only.
Use (c) - Horizontal or vertical.
Use (d) - Any angle between horizontal and vertical.

1.2.4 Hazardous locations. Fittings shall be specified for intended use by locational class and group designations (see 3.1 and 6.2).

1.2.5 Part number. The military specification part number for items described in this specification will be identified as shown in 6.7.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.

STANDARDS

FEDERAL

FED STD H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specification, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

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* UNDERWRITERS' LABORATORIES, INC. (UL)

UL866 - Outlet Boxes and Fittings for Use in Hazardous Locations, Class I, Groups, A, B, C, D, and Class II, Groups E. F. and G.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. This specification establishes the construction and acceptance requirements for explosion proof conduit fittings. Conduit fittings covered by this specification shall comply with the applicable requirements of UL886 covering installation in Division 1. Locations of class I, groups A, B, C, and D; class II, groups E, F, and G; and claass III; as defined in NFPA No. 70 and as summarily defined below (see 1.2.4).

3.1.1 Class I. Locations in which flammable gases or vapors are, or may be, present in the air in quantities sufficient to produce explosive or ignitable mixtures.

3.1.1.1 Group A. Atmospheres containing acetylene.

3.1.1.2 Group B. Atmospheres containing hydrogen, gases, or vapors of equivalent hazard such as manufactured gas.

3.1.1.3 Group C. Atmospheres containing ethyl ether vapors, ethylene, or cyclopropane.

3.1.1.4 Group D. Atmospheres containing gasoline hexane, naptha, benzine, butane, propane, alcohol, acetone, benzol, lacquer solvent vapors, or natural gas.

3.1.2 Class II. Locations which are hazardous because of the presence of combustible dust.

3.1.2.1 Group E. Atmospheres containing metal dust, including aluminum, magnesium and their commercial alloys, and other metals of similarly hazardous characteristics.

3.1.2.2 Group F. Atmospheres containing carbon black, coal, or coke dust.

3.1.2.3 Group G. Atmospheres containing flour, starch, or grain dusts.

3.1.3 Class III. Locations which are hazardous because of the presence of easily ignitable fibers or flyings, but in which such fibers or flying are not likely to be in suspension in quantities sufficient to produce ignitable mixtures.

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* 3.3 Standard commercial product. The conduit fitting shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the conduit fittings being furnished. Standard commercial product is a product which has been or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

* 3.4 Identical items. Conduit fittings of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

* 3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification specifically specified.

3.6 Construction. The construction of conduit fittings shall insure sufficient strength and rigidity in the finished product to enable the product to resist damage from rough handling and during installation and use. They shall be of materials and associated finished commonly used in commercial practices and which meet the qualifying requirements for hazardous locations specified (see 6.2), as well as the requirements of UL886 (see 3.6). Conduit pipe threads shall conform to FED STD H28.

3.7 Fire and casualty hazards.

3.7.1 Each contractor shall submit to the contracting agency proof that the conduit fittings he proposes to supply under this specification conform to the requirements of UL886 (see 3.5). The label or listing of the UL may be accepted as evidence that the fittings conform to this requirement.

3.7.2 In lieu of the label or listing, the bidder may submit a written report satisfactory to the contracting agency stating that his conduit fittings conform to the applicable requirements of the publish standard, including methods of tests, of UL886.

3.7.3 Compliance with the above preliminary requirements in regards to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his materials.

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3.8 Workmanship. Workmanship of the product shall be of the highest quality free from any defects that would not enable the product to conform to the requirements of this specification, or would degrade the appearance of the product, such as dirt, grease, or grim.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein shall be classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Packaging inspection (see 4.6).

4.2.1 First article inspection. First article inspection shall be performed on type, class, and size of fitting when a first article sample is required (see 3.2 and 6.3). This inspection shall include the examination of 4.4.1 and 4.4.2 and the test of 4.6. The first article may be a standard production item from a contractor's current inventory provided the fitting meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining type, class, and size of fitting to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the conduit fittings selected in accordance with 4.3.1. This inspection shall include the examination of 4.4 and the test of 4.5.

4.2.3 Inspection lot. All conduit fittings of the same type, class style and size offered for delivery at one time shall be considered a lot for purpose of inspection.

4.3 Sampling. A random sample of conduit fittings shall be selected from each lot in accordance with MIL-STD-105.

4.3.1 Sampling for examination. Examination of the conduit fittings shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests of the conduit fittings shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

* 4.4 Examination. Each conduit fitting shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

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4.5 Testing. Each sample selected in accordance with 4.3.2 shall be tested in accordance with the applicable tests in UL886. The UL label may be acceptable as evidence of the conduit fittings meeting the tests of the applicable UL. The first article shall be tested in accordance with UL886.

4.6 Packaging inspection. The preservation-packaging, packing, and marking of the conduit fittings shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Unless otherwise specified (see 6.2), the preservation-packaging, packing, and marking of the conduit fitting shall be in accordance with MIL-E-17555. The level of preservation-packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The electrical conduit fittings covered by this specification are intended for use as follows:

Type I. To prevent the passage of gases of combustion and flames from oneportion of the electrical installation to another through conduit.

Type II. To connect equipment where difficult bends are required or where the equipment is subject to vibration or movement.

Type III. To reduce a conduit hub to a smaller size.

Type IV. To connect conduit to a conduit fitting, junction box, or device enclosure.

Type V. To allow accumulated water to drain out of an enclosure and to provide ventilation to minimize condensation in an enclosure.

Type VI. To provide a conduit seal with a removable cover for inspection.

Type VII. To provide a conduit seal with a removable inpsection cover and a drain to prevent accumulation of water above the seal.

Type VIII. To connect conduit to a conduit fitting, junction box, or device enclosure when a turn is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) Size required (see 1.2.2).
(d) Required use (when applicable) (see 1.2.3).
(e) Explosion proof criteria required in accordance with hazardous location classes and groups (see 1.2.4 and 3.6).
(f) When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).

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(g) If preservation-packaging, packing, and marking is other than specified (see 5.1).
(h) Level of preservation-packaging and level of packing required (see 5.1).

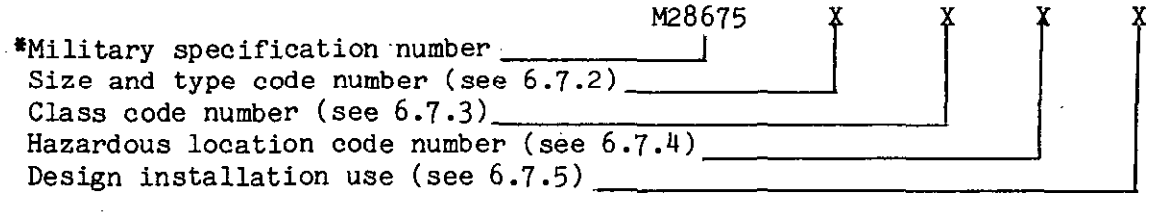
* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item consisting of one complete conduit fitting or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, test, and approval of the first article.

6.4 Special requirements. It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid samples should be definitely stated. The specification shall apply in all other respects. Fittings required for replacements in existing systems wherein dimensional tolerances are critical, shall be specified as to manufacturer (or otherwise detailed) to ensure ease of replacement actions at minimum cost.

6.5 Specification applicability. Military specifications do not necessarily include all types, designs, forms sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.6 Definitive Military Specification part number. The Military Specification part number is a definitive part number which corresponds to the type, class, and size of fittings covered by this specification, and defines the requirements of the options presented under this specification.

6.6.1 Cataloging data. For cataloging purposes, part numbers for the fittings are assigned as follows:



6.6.2 Size and type. The size and type of fitting (see 1.2.1) are identified by a number and letter (see Table VIII).

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TABLE VIII. Code number to size and type.

Fitting size	Type of fittings							
nominal)	Type I	Type II	Type III	Type IV	Type V	Type VI	Type VII	Type VIII
1/8	1A	2A	3A	4A	5A	6A	7A	8A
1/4	1B	2B	3B	4B	5B	6B	7B	8B
3/8	1C	2C	3C	4C	5C	6C	7C	8C
1/2	1D	2D	3D	4D	5D	6D	7D	8D
3/4	1E	2E	3E	4E	5E	6E	7E	8E
1	1F	2F	3F	4F	5F	6F	7F	8F
1-1/4	1G	2G	3G	4G	5G	6G	7G	8G
1-1/2	1H	2H	3H	4H	5H	6H	7H	8H
2	1I	2I	3I	4I	5I	6I	7I	8I
2-1/2	1J	2J	3J	4J	5J	6J	7J	8J
3	1K	2K	3K	4K	5K	6K	7K	8K
3-1/2	1L	2L	3L	4L	5L	6L	7L	8L
4	1M	2M	3M	4M	5M	6M	7M	8M
5	1N	2N	3N	4N	5N	6N	7N	8N
6	10	20	30	40	50	60	70	80

6.6.3 The fitting class (see 1.2.1) is identified by a single letter (see Table IX).

TABLE IX. Code number to class of fitting.

Code number	Fitting class
A	Class 1
B	Class 2
C	Class 3
D	Class 4
E	Class 5
F	Class 6
G	Class 7
H	Class 8
I	Class 9
J	Class 10
K	Class 11
L	Class 12

6.6.4 Hazardous location. Class and group of nazardous location (see 3.1.1 through 3.1.3) shall be identified by a single letter (see Table X).

TABLE X. Code to hazardous location.

Code letter	Class and group of hazardous location
A	Class 1 Group A
B	Class 1 Group B
C	Class 1 Group C
D	Class 1 Group D
E	Class II Group E
F	Class II Group F
G	Class II Group G
H	Class III Group H

6.6.5 Design installation use. The use of the fitting (see 1.2.3) shall be identified by a single digit number (see Table XI).

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TABLE XI. Code to installation use.

Code to number	Installation use
1	Vertical only
2	Horizontal only
3	Horizontal or vertical
4	Any angle between horizontal and vertical

* 6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liabilty whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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